05_gemini_cleanup

March 24, 2026

1 05 Gemini Cleanup and Evaluation

This notebook demonstrates using the Google Gemini API as a late-stage post-OCR correction step. It compares CER/WER before and after cleanup, showing the delta contributed by the LLM correction.

The Gemini prompt is specifically tuned for 17th-century printed Spanish text: it corrects obvious OCR mistakes without modernizing authentic period spellings.

[ ]:
```
import os, sys
from pathlib import Path

cwd = Path.cwd().resolve()
ROOT = cwd if (cwd / 'src').exists() else cwd.parent
if str(ROOT) not in sys.path:
    sys.path.insert(0, str(ROOT))

# Set your API key here or export GEMINI_API_KEY in your shell
os.environ.setdefault('GEMINI_API_KEY', 'YOUR_GEMINI_API_KEY')

from src.postprocess_llm import GeminiCleaner, RuleBasedCleaner
from src.utils import CleanupConfig, read_jsonl
from src.evaluate import compute_cer, compute_wer
```

1.1 1. Test the Gemini connection

[6]:
```
config = CleanupConfig(backend='gemini', gemini_model='models/
↪gemini-flash-lite-latest')
gemini = GeminiCleaner(config)
print(f'Gemini available: {gemini.is_available}')

if gemini.is_available:
    test_text = 'Io, P. GARCIA de Ia COMP. de JESUS'
    cleaned, backend = gemini.clean(test_text)
    print(f'Raw:     {test_text}')
    print(f'Cleaned: {cleaned}')
    print(f'Backend: {backend}')
```

```
Gemini available: True
Raw:     Io, P. GARCIA de Ia COMP. de JESUS
Cleaned: Io. P. GARCIA de la COMP. de JESUS
Backend: gemini
```

1.2 2. Compare rule-based vs Gemini cleanup on baseline predictions

Load the predictions file written by run_baseline.py and re-run both cleaners.

[7]:
```
PREDICTIONS_FILE = ROOT / 'data' / 'predictions' / 'baseline_predictions.jsonl'
GROUND_TRUTH_FILE = ROOT / 'data' / 'ground_truth' / 'ground_truth.jsonl'

if not PREDICTIONS_FILE.exists():
    print('No predictions file. Run run_baseline.py first.')
else:
    predictions = read_jsonl(PREDICTIONS_FILE)
    ground_truth = {r['page_id']: r['text'] for r in␣
 ↪read_jsonl(GROUND_TRUTH_FILE)}
    rule_cleaner = RuleBasedCleaner(config)

    results = []
    for pred in predictions:
        page_id = pred['page_id']
        if page_id not in ground_truth:
            continue
        ref = ground_truth[page_id]
        raw = pred.get('raw_ocr', '')
        rule_cleaned = rule_cleaner.clean(raw)

        if gemini.is_available:
            gemini_cleaned, _ = gemini.clean(raw)
        else:
            gemini_cleaned = rule_cleaned

        results.append({
            'page_id': page_id,
            'raw_cer': compute_cer(ref, raw),
            'rule_cer': compute_cer(ref, rule_cleaned),
            'gemini_cer': compute_cer(ref, gemini_cleaned),
            'raw_wer': compute_wer(ref, raw),
            'rule_wer': compute_wer(ref, rule_cleaned),
            'gemini_wer': compute_wer(ref, gemini_cleaned),
        })
        print(f"{page_id}: raw_CER={results[-1]['raw_cer']:.4f} ␣
 ↪rule_CER={results[-1]['rule_cer']:.4f} ␣
 ↪gemini_CER={results[-1]['gemini_cer']:.4f}")

    if results:
```

```
    avg = lambda key: sum(r[key] for r in results) / len(results)
    print(f"\nAverage CER - Raw: {avg('raw_cer'):.4f}  Rule:␣
↪{avg('rule_cer'):.4f}  Gemini: {avg('gemini_cer'):.4f}")
    print(f"Average WER - Raw: {avg('raw_wer'):.4f}  Rule: {avg('rule_wer'):
↪.4f}  Gemini: {avg('gemini_wer'):.4f}")
```

```
buendia_instruccion_page_0001: raw_CER=0.9480  rule_CER=0.9463
gemini_CER=0.8667
buendia_instruccion_page_0002: raw_CER=0.9181  rule_CER=0.9209
gemini_CER=0.9473
buendia_instruccion_page_0003: raw_CER=1.5745  rule_CER=1.5597
gemini_CER=1.5365

Average CER - Raw: 1.1468  Rule: 1.1423  Gemini: 1.1168
Average WER - Raw: 1.3387  Rule: 1.3050  Gemini: 1.2974
```

1.3 3. Side-by-side example

Pick a single page and display the raw OCR, rule-based cleanup, and Gemini cleanup side by side.

[8]:
```
if results:
    # Use the page with the highest raw CER for a dramatic comparison
    worst = max(results, key=lambda r: r['raw_cer'])
    pred = next(p for p in predictions if p['page_id'] == worst['page_id'])
    ref = ground_truth[worst['page_id']]
    raw = pred.get('raw_ocr', '')

    print(f"=== Page: {worst['page_id']} ===")
    print(f"Raw CER {worst['raw_cer']:.4f}  ->  Gemini CER {worst['gemini_cer']:
↪.4f}")
    print()
    print('--- GROUND TRUTH (first 400 chars) ---')
    print(ref[:400])
    print()
    print('--- RAW OCR ---')
    print(raw[:400])
    print()
    gemini_cleaned, _ = gemini.clean(raw) if gemini.is_available else␣
↪(rule_cleaner.clean(raw), 'rule')
    print('--- GEMINI CLEANED ---')
    print(gemini_cleaned[:400])
```

```
=== Page: buendia_instruccion_page_0003 ===
Raw CER 1.5745  ->  Gemini CER 1.5365

--- GROUND TRUTH (first 400 chars) ---
crianza de la niñez. Assi sea,
Divinissimo Niño, por vuestra
```

gracia, assi sea, a vuestra ma-
yor gloria. Amen.
CENSURA DEL R. P. ANTONIO CO-
dorniu de la Compañía de jesus, Maes-
tro que fue de Theologia, Examinador
Synodal de los Obispados de Gerona, Ur-
gel, y Barcelona, Oc.
DE orden del Ilustre Señor Don Fran-
cisco Drechos, Canonigo, y Sacristan
Dignidad de la Santa Iglesia de Gerona, y
Vicari

--- RAW OCR ---
WWW.LLM
GARDORIZIO ON 6020 : LAKEL# : 1 1 1
GOOD DARACON DIOS ON IA DEVORA LAVINAS BENDICONES, LES UNILLEIS.
MASTERIA LOS TEMPLOSJA PEDAD A VEGERO BARADO PECHO ONA-
COM LOS PADIES EN LA OBEDIENIA, WILSIMOS ADRAOS, DICHOTA EDAD, M
ITEMS INCLUDS; #2 INDOCONA, F OR# QPC OS MOTOCO GAN REGALADOS CR- "
TEO DE LABER Y COM IOS MAYORES, MISS!
ENTANGO MAS OF OUR, I PERSON. Y DOES ON A ORIGINAL IETA-
NET TOT

--- GEMINI CLEANED ---
GARDORIZIO ON GOD: LAKEL: :
BUEN DARACON DIOS EN LA DEVORA LAMINAS BENDICIONES, LES UNIMOS.
MAIESTAR LOS TEMPLOS JA PEDID A VEGERO BARADO PECHO ONA-
COM LOS PADRES EN LA OBEDIENCIA, HUMILISIMOS ABRAZOS, DICHOSA EDAD, M
ITEMS INCLUSOS: INDOCINA, FORQUE OS MUESTRO GOS REGALADOS CR-"
TEO DE LABER Y COM IOS MAYORES, MIS
ENTANTO MAS DE NOS, PERSONA, Y DE LOS ORIGINAL LETRA-
NET TOTAL AMOUNT
*** TAX. UN